

global warming, the phenomenon of increasing average [air temperatures](#) near the surface of [Earth](#) over the past one to two centuries. Climate scientists have since the mid-20th century gathered detailed observations of various [weather](#) phenomena (such as temperatures, [precipitation](#), and storms) and of related influences on [climate](#) (such as [ocean currents](#) and the atmosphere's chemical composition). These data indicate that Earth's climate has changed over almost every conceivable timescale since the beginning of geologic time and that [human](#) activities since at least the beginning of the [Industrial Revolution](#) have a growing influence over the pace and extent of present-day [climate change](#).

Giving voice to a growing [conviction](#) of most of the scientific [community](#), the [Intergovernmental Panel on Climate Change](#) (IPCC) was formed in 1988 by the [World Meteorological Organization](#) (WMO) and the [United Nations Environment Program](#) (UNEP). The IPCC's Sixth [Assessment Report](#) (AR6), published in 2021, noted that the best estimate of the increase in global average surface temperature between 1850 and 2019 was 1.07 °C (1.9 °F). An IPCC special report produced in 2018 noted that human beings and their activities have been responsible for a worldwide average temperature increase between 0.8 and 1.2 °C (1.4 and 2.2 °F) since preindustrial times, and most of the warming over the second half of the 20th century could be attributed to human activities (*see also* [global warming policy](#)).

AR6 produced a series of global climate [predictions](#) based on modeling five greenhouse gas emission scenarios that accounted for future emissions, mitigation (severity reduction) measures, and uncertainties in the model projections. Some of the main uncertainties include the precise role of feedback processes and the impacts of industrial pollutants known as

[aerosols](#), which may offset some warming. The lowest-emissions scenario, which assumed steep cuts in greenhouse gas emissions beginning in 2015, predicted that the global mean surface temperature would increase between 1.0 and 1.8 °C (1.8 and 3.2 °F) by 2100 relative to the 1850–1900 average. This range stood in stark contrast to the highest-emissions scenario, which predicted that the mean surface temperature would rise between 3.3 and 5.7 °C (5.9 and 10.2 °F) by 2100 based on the assumption that greenhouse gas emissions would continue to increase throughout the 21st century. The intermediate-emissions scenario, which assumed that emissions would stabilize by 2050 before declining gradually, projected an increase of between 2.1 and 3.5 °C (3.8 and 6.3 °F) by 2100.

Many climate scientists agree that significant societal, economic, and ecological damage would result if the global average temperature rose by more than 2 °C (3.6 °F) in such a short time. Such damage would include increased [extinction](#) of many plant and animal species, shifts in patterns of [agriculture](#), and rising [sea levels](#). By 2015 all but a few national governments had begun the process of instituting carbon reduction plans as part of the [Paris Agreement](#), a treaty designed to help countries keep global warming to 1.5 °C (2.7 °F) above preindustrial levels in order to avoid the worst of the predicted effects. Whereas authors of the 2018 special report noted that should carbon emissions continue at their present rate, the increase in average near-surface air temperature would reach 1.5 °C sometime between 2030 and 2052, authors of the AR6 report suggested that this [threshold](#) would be reached by 2041 at the latest. In an ominous sign that the long-term temperature threshold of 1.5 °C could arrive far sooner than expected, global climate monitoring databases reported that the global temperature average for the years 2024 and 2025 were 1.6 °C and 1.4 °C above preindustrial levels, respectively.

